CHAPTER 22

Behavior of Radionuclides in the Environment

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The main objection against nuclear power is the risk of spread of "radioactivity" (radioactive elements) to the environment where it may cause health effects in humans. Such effects have already been discussed in chapter 15. As mentioned previously there are several suggested methods for the treatment of used nuclear fuel. However, all of them will require a final repository for the waste. The main difference is the total storage time, the different barriers, and the radiotoxicity of the final waste. In any case, if the repository is breached and ground water comes in contact with the fuel the readionuclides will be dissolved to a smaller or greater extent, depending on element and form. In this chapter, the chemical aspects of the sources of releases, and of the migration of the radionuclides in the environment, will be discussed. Their chemical properties, together with hydrology, determine how fast they will move from their point of entry into the

Radiochemistry and Nuclear Chemistry
ISBN 978-0-12-405897-2, http://dx.doi.org/10.1016/B978-0-12-405897-2.00022-7
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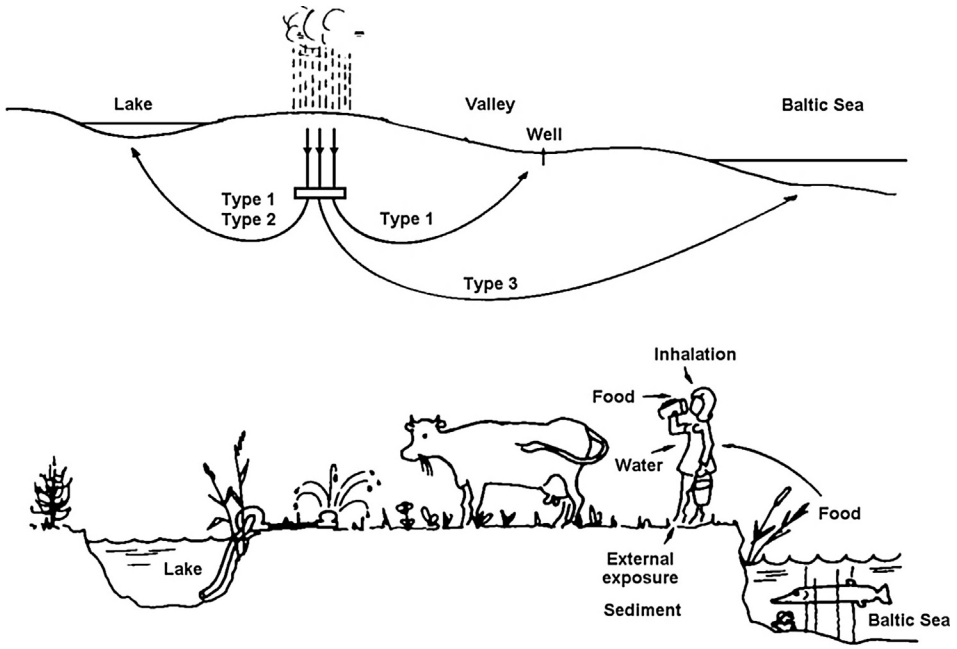


Figure 22.1 Migration path of radioactive nuclides from a waste repository to man.

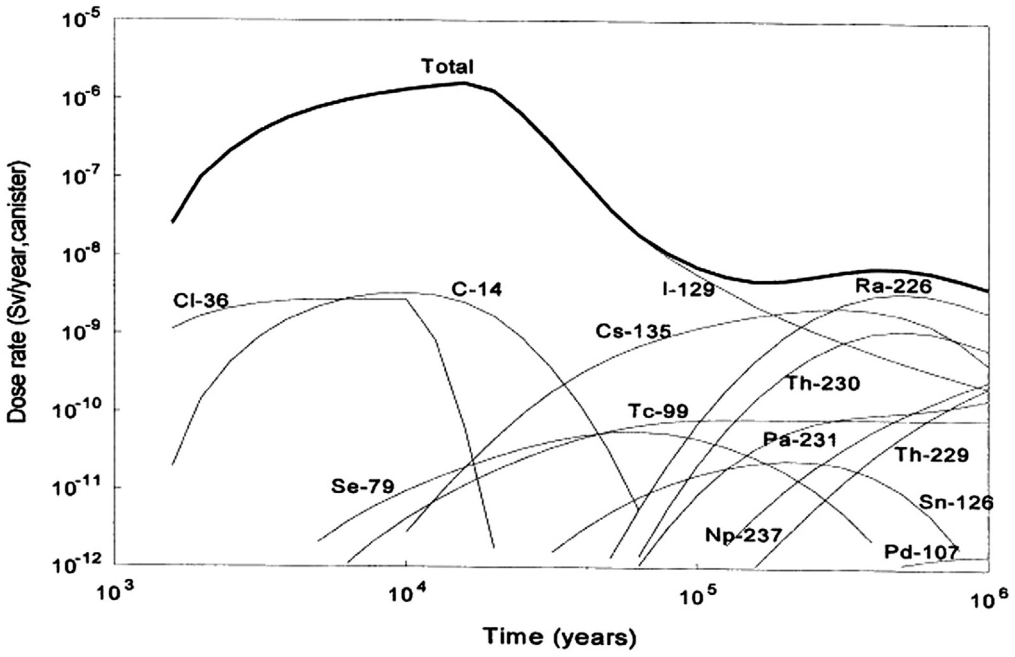


Figure 22.2 Predicted dose rates to individuals from the release of a defective canister, assuming the release occurs directly to the biosphere, i.e. without retention in the ground (*From SKI Site97*).

groundwater to water resources used by man; this is schematically illustrated in Figure 22.1. In particular we discuss actinide behavior as these elements have the most hazardous radionuclides which may be released in the different steps of the nuclear fuel cycle, and, especially, from nuclear waste repositories. However, despite this it may be noted here that the actinides contribution to the dose from a leaking cannister to the recipient is negligible. This is exemplified by the dose to a recipient from one leaking cannsiter in the planned Swedish repository shown in Figure 22.2.

22.1. RADIOACTIVE RELEASES AND POSSIBLE EFFECTS

In earlier chapters, there have been brief discussions of the release of radionuclides to the environment. Such releases occur from mining and milling operations, particularly of uranium ores (§14.2.4.4), from the nuclear fuel fabrication processes, from normal operation of nuclear reactors (§19.16), from reprocessing of spent nuclear fuel (§21.8), from nuclear weapons production and recovery, from transportation of nuclear material, from testing of nuclear weapons and accidents (§13.10), and from storage of nuclear wastes (which is discussed in subsequent sections). Table 22.1 compares the estimated collective dose to the public from the various activities of the nuclear fuel cycle, from

Table 22.1 Collective effective dose to the public from radionuclides released in effluents from the nuclear fuel cycle 1995-97 (UNSCEAR 2000)

Source	Normalized collective effective dose (manSv/GW_ey)
	Local and regional effects
Mining	0.19
Milling	0.008
Mine and mill tailings (releases over 5 years)	0.04
Fuel fabrication	0.003
Reactor operation	
Atmospheric	0.4
Aquatic	0.04
Reprocessing	
Atmospheric	0.04
Aquatic	0.09
Transportation	<0.1
	Solid waste disposal and global effects
Mine and mill tailings	
Releases of radon over 10 000 y	7.5
Reactor operation	
Low-level waste disposal	0.00005
Intermediate-level waste disposal	0.5
Reprocessing solid waste disposal	0.05
Globally dispersed radionuclides (truncated to 10 000 years)	40